Massey University
College of Sciences

161.304: Computer Laboratory 1

Tips and Comments on R Syntax

- R is case sensitive (so `LM()` and `lm()` are not equivalent).
- R is tolerant to the use of spaces, so `x <- 1` and `x<-1` are equivalent. I try to put spaces to make the code readable.
- You can use the arrow keys on your keyboard to speed things up. The 'up' arrow gives you the previous command that you typed.
- The usual prompt sign for R is `>`. If you get a `+` prompt sign instead, it means that R is awaiting the completion of the previous command that you typed in. This can happen because you have forgotten to close parentheses, for instance. Just type in the remainder of the command.

Exercise 1: R as a Calculator

Use R to compute the following:

1. $98765 - 43210$
2. 12.3456789^2
3. $\sqrt{(1.23 + 4.56) \times (7.8 + 9.0)}$

Type the following expressions into R, and check that you understand the results.

1. `3+4*5`
2. `3/4/5`
3. `1:20/2`
4. `(1:20)/2`

Exercise 2: Exploratory Analysis of IBM Stock Price Data

This exercise is concerned with the closing price (in U.S. dollars) of IBM stock over a sequence of 250 consecutive trading days in 1980.

These data can then be read in and assigned to the vector `ibm`, with the following command

```
ibm <- scan(file="https://raw.githubusercontent.com/oliviaAB/161304/master/ibm.txt")
```

If you cannot load the data directly in this way, then download them from the 161.304 Stream website and save them on your computer. The data can then be read into R by:

```
ibm <- scan(file="<file path>/ibm.txt")
```

where you should replace `<file path>` with the appropriate 'address' corresponding to the location where you stored the file on your computer.

1. Take a look at the data by typing the name of the vector: `ibm`.

2. Find the mean, standard deviation, maximum and minimum of the IBM stock price data.

3. You should have found in question 2 that the maximum price is the (rather unlikely) value 6575. This is, in fact, an outlier generated by a typographical error. Find out which element of `ibm` corresponds to this outlier, and replace that element by the correct value which is 65.75.

4. Recalculate the summary statistics from question 2 using the corrected data set.

5. The *return* on any given day is defined to be the stock price on that day divided by the stock price on the previous day. Create a vector containing the returns.

6. It is often assumed in financial modelling that the log-returns are (approximately) independent and normally distributed. Create a vector containing the logarithms of the daily returns; call it `lreturn`.

7. Investigate whether the normality assumption seems reasonable by producing a histogram of the data with the command `hist(lreturn)`. Try `hist(lreturn,breaks=15)` to get a finer level of discretization.

8. Type `t.test(lreturn,mu=0)` and interpret the results. Find a 99% confidence interval for the (population) mean log-return by `t.test(lreturn,mu=0,conf.level=0.99)`.

Exercise 3: Professor Kolmogorov Plays Cricket

Professor Kolmogorov is a keen cricketer. He has noted that his batting performances tend to go through periods of good and bad form, and so decides to model them using a Markov chain. Specifically, he categorises each of his innings as follows:

Category	Definition
1	Disaster: made 5 runs or less.
2	Poor: made between 6 and 15 runs.
3	Adequate: made between 16 and 29 runs.
4	Good: made between 30 and 49 runs.
5	Excellent: made 50 or more runs.

The transition matrix for this chain is

$$P = \begin{bmatrix} 0.4 & 0.2 & 0.2 & 0.1 & 0.1 \\ 0.3 & 0.3 & 0.2 & 0.1 & 0.1 \\ 0.2 & 0.2 & 0.3 & 0.2 & 0.1 \\ 0.1 & 0.2 & 0.3 & 0.2 & 0.2 \\ 0.1 & 0.1 & 0.2 & 0.4 & 0.2 \end{bmatrix}$$

1. Create a matrix `P` in R for the transition matrix.

2. Remember that the transition matrix is stochastic: i.e. the row sums are all one. Check that this is the case with the command

   ```
   rowSums(P)
   ```

   If any of the row sums are not equal to one, then you have made an error when transcribing P. Find and correct it.

3. Professor Kolmogorov starts the season in disastrous form; i.e. $Y_0 = 1$. Find the probability that he is in excellent form at innings $t = 5$ (i.e. find $P(Y_5 = 5|Y_0 = 1)$).

4. In innings $t = 10$, Professor Kolmogorov scores 20 runs. Find the probability that he will score 30 or more runs in innings $t = 13$.

Exercise 4: Simulating Professor Kolmogorov's Cricket Season

Assuming that he starts in disastrous form, so $Y_0 = 1$, write R code to simulate Professor Kolmogorov's batting performance over the next $n = 20$ innings using the Markov chain model from Exercise 1. Produce a plot of y_t against t to display the results.

For how innings many of these 20 innings can Professor Kolmogorov expect to be excellent? To answer this question you will need to produce a large number of different simulations of the season, and then look at the average results. Specifically, to create 1000 separate simulations, you will need to write a *double loop*, of the form

```
for ( i in 1:1000 ){
.....
      for (tt in 1:20){
      .....
      }
results[i] <- .....
}
```

where `results` is a vector of length 1000 for which the ith element will store the number of excellent innings in the ith simulation.

The first row of dots will need to be replaced by any necessary set up for each inner loop. For the final set of dots (assigning a value of `results[i]`), you will need to extract from a vector of simulated states the number which are equal to 5. Suppose that the simulations are held in a vector `y`. Then `sum(y==5)` returns the number of elements of `y` that are equal to 5. To understand why, recall that `y==5` is a logical comparison, and so will return a sequence of `T` (for each element of `y` that equals 5) and `F` (for each element that does not). When applying a numerical function like `sum`, R coerces the logical vector into a binary vector (1 replaces `T` and 0 replaces `F`) and so the sum of these is the number of elements of `y` that equal 5.

Consult the Comments and Solutions sheet if you get completely stuck, but make sure that you have a good try at figuring out your own solution before doing so.

Finally, note that there was no compulsion for me to indent the code for the inner loop in the R syntax given above. Nonetheless, many programmers find it helpful to do so in order to keep track of the structure of their code.